WordWreck Code Defense Guide

Checked against the current Java source at commit a569e77 (14 October 2026). Search by method name if code moves later.

What runs the game?

GameEngine is the Java-only rule book. **PrimaryController** connects keys, buttons, dictionary approval, and the JavaFX screen to that engine. **Levels.ALL** provides the 20 fixed answers. The separate **backend** package is not called by the active game.

Find this	Purpose
preview/GameEngine.java	Validates guesses, marks letters, secures the anchor, chooses outcomes and scores.
preview/Levels.java and Level.java	Store answers, clues, and the two positions of the shared letter.
PrimaryController.java	Receives keys, approves input, calls the engine, and redraws the board.
DictionaryService.java	Checks complete guesses in dictionary mode; cannot spend an attempt.
App.java, primary.fxml, CSS	Start and display the JavaFX interface.
GameSettings.java	Persists options and best scores using Java Preferences.

The Level 1 example to remember

A is **CARGO**; B is **SHARKS**. Their crossing letter is **R**: index 2 of CARGO (third letter) and index 3 of SHARKS (fourth letter). Java indexes start at 0. The screen starts with A. B opens when A is solved, or after five unsuccessful A guesses if at least one guess made that R green.

For example, **PARTY** is not CARGO, but its third letter R secures A's anchor. If A still is not solved after five guesses, B opens with its R already filled.

Walk through GameEngine.submitGuess: input to attempt

Open **demo/src/main/java/com/example/preview/GameEngine.java** and search **submitGuess(String input)**. The line ranges below match the current file. The controller calls this method only after the typed word is approved.

Current lines	Plain explanation with Level 1
93-96	If the outcome is no longer PLAYING, reject another guess. A finished level cannot gain an attempt.
98-103	Make **guessedWord**. Null becomes empty; otherwise trim spaces and uppercase the input. For example, ' cargo ' becomes CARGO.
104-107	**currentAnswer()** chooses CARGO on A or SHARKS on B. The regular expression requires exactly that many A-Z letters. 'CAT' is too short for A, so an exception stops the method.
108-111	Only on B, compare the guess's anchor position with B's fixed anchor letter. For SHARKS, index 3 must be R. SHALKS has L there, so it is rejected with 'letter 4 must be R'.
113	Call **evaluateLetters(answer, guessedWord)** to calculate tile marks; package the word and marks in a **Guess**.
114-120	Choose A's or B's history and add the Guess. **This is where one attempt is recorded.** All earlier rejected input stopped before this line.
121	Compare the entire guess with the answer. **guessedCorrectAnswer** is true only for an exact word match.

What do the arguments mean?

input is the String sent by PrimaryController from the current tiles. **answer** is the hidden target. **guessedWord** is the normalized player guess. The two arguments in **evaluateLetters(answer, guessedWord)** are passed in that order: target first, guess second.

anchorB() is passed into **charAt(index)** to select B's shared letter. In Level 1, index 3 selects R. The **IllegalArgumentException** argument is the message shown to the player; adding 1 turns Java index 3 into player position 4.

Walk through GameEngine.submitGuess: decide the result

Current lines	Plain explanation with Level 1
123-125	If still on A, keep the anchor secured if the new guess made A's R green. **|=** means an earlier green anchor stays secured even when a later guess misses.
126-128	If A is exactly CARGO, mark A solved and switch to B. The correct guess is checked before a fifth-guess failure.
129-132	If A is still wrong after its fifth recorded guess, open B only when the anchor was secured; otherwise set IMMEDIATE_SINK.
133-140	If on B and the word is exactly SHARKS, mark B solved. Choose FULL_VICTORY if A was solved; otherwise SURVIVAL_CLEAR.
141-148	If B is still wrong after its fifth guess, choose PARTIAL_SURVIVAL if A was solved; otherwise BOTH_SUNK.
149	Return the Guess. Its word and marks are available to a caller. The current controller ignores the return value and redraws from the updated engine state.

All five final outcomes

A	B	Outcome	Score
Solved	Solved	FULL_VICTORY	100%
Solved	Five misses	PARTIAL_SURVIVAL	50%
Five misses, anchor secured	Solved	SURVIVAL_CLEAR	50%
Five misses, anchor secured	Five misses	BOTH_SUNK	0%
Five misses, no anchor	Never reached	IMMEDIATE_SINK	0%

scorePercent() maps the stored final Outcome to 100, 50, or 0. It does not calculate a percentage from the number of guesses. During PLAYING it returns 0, but the final score screen is only shown after a final outcome.

How tile colors are calculated

Search **GameEngine.evaluateLetters(answer, guessedWord)**. It returns one Mark for each position: **CORRECT** (green, right place), **PRESENT** (gold, elsewhere), or **ABSENT** (gray, no unused matching copy). This method does not choose A/B outcomes.

Step	What happens
Length check	Answer and guess must have the same length. submitGuess already checks this, but the method protects itself when called directly.
Start gray	Create a Mark array and fill every position with ABSENT.
Pass 1	Mark letters at the correct positions green. Count the answer letters not used by those green matches.
Pass 2	For each non-green guessed letter, mark it gold only when an unused matching answer letter remains. Reduce that remaining count by one.
Return	Return an ordered, unmodifiable List of marks.

Actual Level 1 example: answer CARGO, guess RACER

CARGO has C-A-R-G-O. RACER has R-A-C-E-R. A at position 2 is green. The first R and C are gold because those letters exist elsewhere in CARGO. E is gray. The final R is gray because CARGO has only one R and the earlier R already used it. This is why the algorithm makes two passes and tracks remaining counts.

Why the shark does not advance on a correct final guess

wrongGuessCountForWord(wordB) starts with that word's number of recorded guesses. If the word was solved, it subtracts one: the last guess was correct. The shark uses this wrong-guess count. **wordB=false** selects A; **wordB=true** selects B.

Locked anchor versus secured anchor

anchorSecured records whether any A guess made the crossing tile green. After B opens, PrimaryController fills B's anchor letter automatically and skips it during typing and deletion. GameEngine also rejects a B guess that changes the letter, so the rule holds even if the engine is called directly.

From key press to game result

Search in PrimaryController	What to say
beginNewLevel	Gets a Level from Levels.ALL, creates GameEngine, resets typed input, and shows the game.
isTypingAllowed	Only allows edits while a game exists, the game pane is visible, and Outcome is PLAYING.
isLockedAnchorPosition	On B, identifies the anchor slot so typing and Backspace skip it.
resetTypedLettersForCurrentWord	Creates the current word's char array and fills B's anchor automatically.
typeLetterIntoFirstEmptyTile / eraseLastTypedLetter	Fill the first empty editable tile or clear the last filled editable tile, then revalidate.
revalidateTypedWord	Controls offline approval or asynchronous dictionary lookup. A stale lookup cannot approve edited letters.
submitGuess	Checks full input and approval; then calls game.submitGuess(new String(typedLetters)).
refreshGameDisplay / showFinalResult	Redraw the board and show the final outcome, sound, and saved best score.

Dictionary mode does not spend attempts by itself

Search **DictionaryService.validateWord**. Offline Practice accepts a complete 5-9 letter A-Z guess immediately without an HTTP request. With Dictionary ON, curated level answers are cached; other words go to Datamuse, then Free Dictionary if Datamuse is unavailable, then the bundled **words.txt** list if online checks cannot answer. A rejected result leaves Enter blocked, so the engine is not called and no attempt is spent.

lookupRequestVersion changes when letters or mode change. The later callback checks this version and the exact typed String before enabling Enter. This prevents an old dictionary reply from approving new letters.

Quick defense questions and answers

Likely question	Answer and method to show
What is the main logic?	GameEngine.submitGuess updates attempts, anchor, and Outcome; evaluateLetters computes colors; scorePercent maps Outcome to score.
Can B open if A is wrong?	Yes. After five unsuccessful A guesses, a green A anchor opens B. Show anchorSecured and the A branch in submitGuess.
What does the B anchor check do?	It rejects a B guess whose shared letter differs. Level 1 B must have R as its fourth letter. The exception happens before activeHistory.add(guess).
What is Outcome.PLAYING?	Outcome is an enum of whole-level states. PLAYING means the round is ongoing; outcome stores the current enum value.
Why use Locale.ROOT?	Uppercasing a guess consistently, independent of the computer's language setting.
What is JavaFX here?	App loads primary.fxml and starts the window. The controller reacts to input and draws the UI. GameEngine contains the rules without JavaFX.
Where are best scores?	GameSettings stores separate practice and dictionary best scores with Java Preferences and never replaces a best score with a lower one.

Useful parameter meanings

index / levelIndex: zero-based position in Levels.ALL; 0 means Level 1. **anchorA / anchorB:** zero-based positions of the same crossing letter. **input:** proposed guess String. **answer:** current hidden word. **wordB / down:** false means A/across, true means B/down. **typedLetters:** char array on the screen; zero means an empty tile. **Mark:** feedback for one letter. **Outcome:** status of the whole level.

During the defense, search the method name in the editor. If asked about a line, say: what condition it checks, what value or state it changes, and what the player sees as a result.